DVD Rental Analysis Report

Introduction

This project analyzes customer rental data from a DVD rental database using SQL. The dataset consists of transactional and description data related to films, customers, rentals, inventory and payments.

The objective of this analysis is to identify key rental patterns, customer behavior, and revenue trends. By leveraging SQL queries and relational joins, the project aims to generate insights that can support business decisions such as inventory management, customer retention, and revenue optimization.

Data & Methodology

The dataset was sourced from Kaggle and includes five main tables:

- Film
- Inventory
- Rental
- Customer
- Payment

These tables were imported and managed using PostgreSQL (pgAdmin 4).

To perform the analysis, INNER JOIN operations were used to combine related tables through their primary and foreign keys. Aggregation functions such as SUM(), COUNT(), and AVG() were applied to derive meaningful metrics.

Business Questions and Analysis

Question 1: What is the total revenue generated?

Method:
To determine total revenue, the Payment table is used. The SUM() function is applied to the amount column:

```sql
-- Q1: Total revenue
SELECT SUM(amount) AS total_revenue
FROM payment;
```

Results:

	total_revenue numeric
1	67416.51

Insights:

The total revenue generated from rentals is $67 416.51, indicating the overall financial performance of the DVD rental business within the database period.

This figure provides a baseline for evaluating business success and can be used to track growth over time.

Understanding total revenues also helps identify whether improvements in customer retention or pricing strategies are needed to increase profitability.

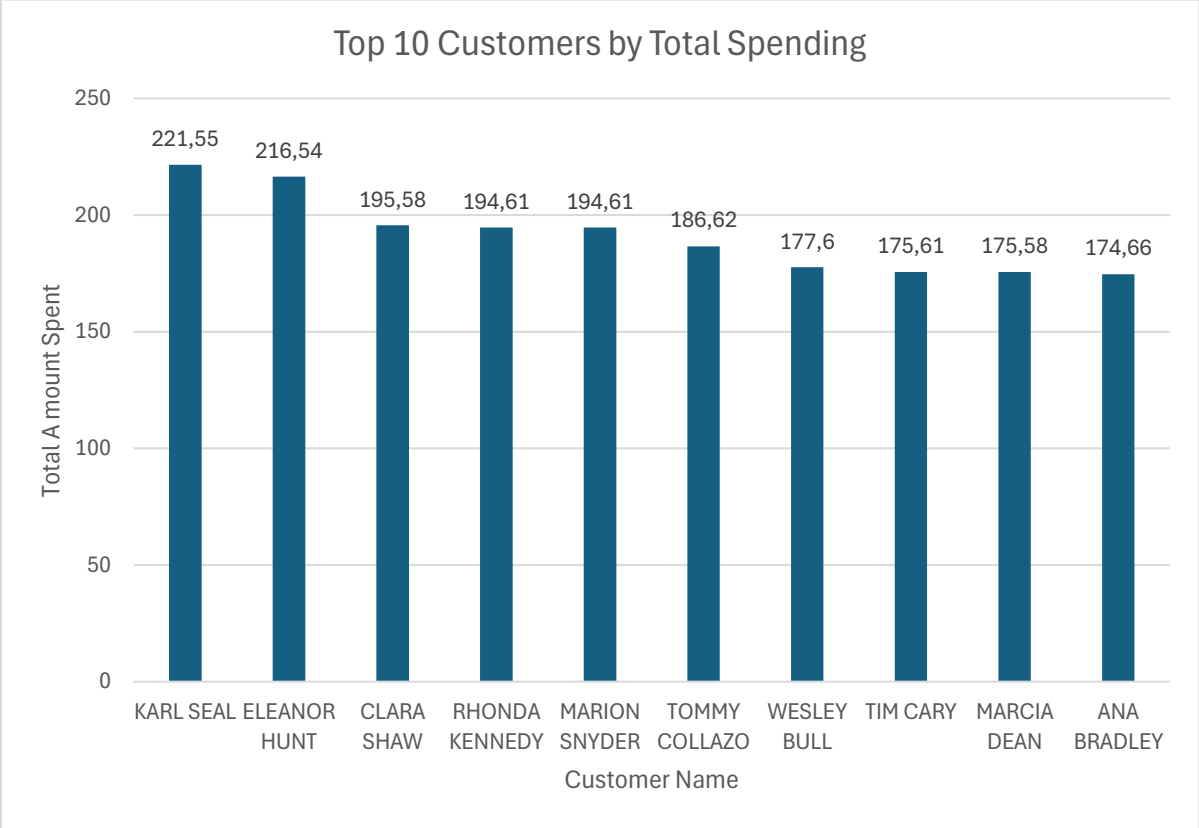
Question 2: Who are the top customers?

Method:
The Customer and Payment tables are joined. The CONCAT function is used to combine first and last names.
Total spending per customer is calculated using SUM(), grouped using GROUP BY, and sorted in descending order.

```
-- Q2: Top 10 customers

SELECT CONCAT(c.first_name,' ',c.last_name) AS customer_fullname, SUM(amount) AS total_spent
FROM payment AS p
INNER JOIN customer AS c ON p.customer_id = c.customer_id
GROUP BY CONCAT(c.first_name,' ',c.last_name)
ORDER BY total_spent DESC
LIMIT 10;
```

Results:



Insight:
The top 10 customers account for total spending ranging from $174.66 to $221.55, with Karl Seal leading at $221.55.

The relatively narrow spending range among the top customers suggests a balanced distribution of high value customers, rather than reliance on a single dominant spender.

This indicates that customer loyalty is spread across multiple individuals, making retention strategies more stable and less risky.

The business should focus on maintaining engagement with this group through targeted promotions or loyalty programs.

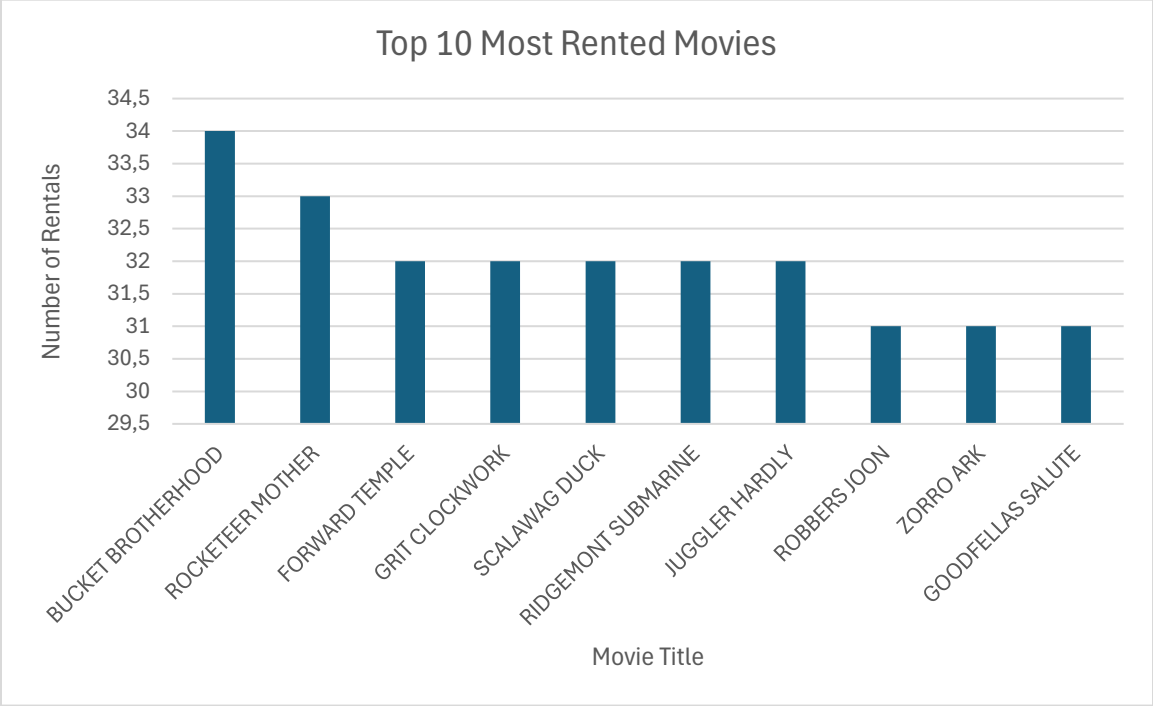
Question 3: Which are the most rented movies?

Method:
The Inventory and Film tables are joined. Rental frequency is calculated by counting rentals per movie.

```sql
-- Q3: Most rented movies

SELECT f.title, COUNT(*) AS times_rented
FROM film AS f
JOIN inventory AS i ON f.film_id = i.film_id
JOIN rental AS r ON i.inventory_id = r.inventory_id
GROUP BY f.title
ORDER BY times_rented DESC
LIMIT 10;
```

Results:



Insight:
The most rented movie is Bucket Brotherhood with 34 rentals, followed closely by other top-performing titles with similar rental frequencies.

The small difference of only 8 rentals between the top movies suggests consistent demand across multiple films, rather than a single dominant title.

This indicates a diversified customer preference, reducing dependency on one blockbuster.

From business perspective, maintaining sufficient inventory across these top titles is essential to meet demand and avoid stock shortages.

Question 4: What is the total revenue per movie?

Method:
The Film, Inventory, Rental, and Payment tables are joined. SUM() is used to calculate revenue per movie.

```sql
-- Q4: Revenue per movie

SELECT f.title, SUM(p.amount) AS revenue
FROM film AS f
INNER JOIN inventory AS i ON f.film_id = i.film_id
INNER JOIN rental AS r ON i.inventory_id = r.inventory_id
INNER JOIN payment AS p ON r.rental_id = p.rental_id
GROUP BY f.title
ORDER BY revenue DESC
LIMIT 10;
```

Results:

	title text	revenue numeric
1	TELEGRAPH VOYAGE	231.73
2	WIFE TURN	223.69
3	ZORRO ARK	214.69
4	GOODFELLAS SALU...	209.69
5	SATURDAY LAMBS	204.72
6	TITANS JERK	201.71
7	TORQUE BOUND	198.72
8	HARRY IDAHO	195.70
9	INNOCENT USUAL	191.74
10	HUSTLER PARTY	190.78

Insight:
The revenue per movie analysis shows that certain films generate significantly higher income than others.

This indicates that high revenue films are not only frequently rented but may also benefit from higher pricing or longer rental duration.

Identifying these top-performing films allows the business to prioritize them in the marketing campaigns and ensure consistent availability.

Focusing on these revenue-driving titles can directly contribute to increased overall profitability.

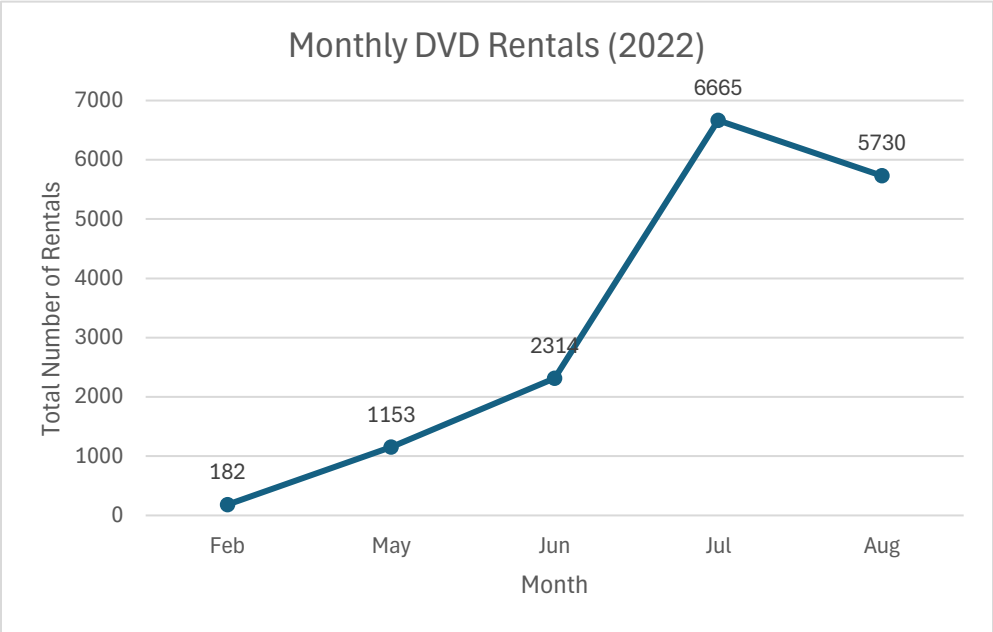
Question 5: How many rentals occur over a monthly period?

Method:
Rental dates are grouped by month, and total rentals are counted.

```
-- Q5:Rentals over time

SELECT DATE_TRUNC('month', rental_date) AS rental_month,
TO_CHAR(DATE_TRUNC('month', rental_date), 'Mon') AS month_name,
COUNT(*) AS total_rentals
FROM rental
GROUP BY rental_month
ORDER BY rental_month;
```

Result:



Insight:

Rental activity shows a noticeable increase during the mid-year period, particularly between May and July, followed by a decline thereafter. This pattern suggests a seasonal trend in customer behavior.

The increase in rentals during these months may be associated with colder seasons, where customers are more likely to stay indoors and engage in home entertainment.

The absence of data for certain months may indicate missing records or periods of low activity, which should be considered when interpreting the results.

From a business perspective, aligning promotions, inventory planning, and marketing efforts with peak rental periods can help maximize revenue opportunities.

Conclusion

This analysis provides valuable insights into customer behavior, movie performance, and revenue generation within the DVD rental business.

Key findings include:

- A stable group of high-value customers contributing to revenue
- Consistent demand across multiple popular films
- Seasonal trends influencing rental activity

These insights can support data-driven decision-making in areas such as customer retention, inventory optimization, and marketing strategy.

Overall, this project demonstrates how SQL can be effectively used to transform raw data into meaningful business insights.